10.29 Gamma-Matrix Traces

LecNote17 pp. 10–12 (Srednicki §47, taught in Class 25 with LecNote20) list the trace identities and sketch their proofs. LecNote20 pp. 4–6 start the computation they serve, the spin-summed $|T|^2$ for $e^-\varphi \to e^-\varphi$ in Yukawa theory, and stop at "only two types". The questions:

- "I follow the arrow on LecNote17 p. 11 and get the wrong signs. How does the four-γ trace come out?"
- "Which sign of ϵ^{0123} goes with $\mathrm{Tr}[\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -4i\epsilon^{\mu\nu\rho\sigma}$?"
- "Another book has $\gamma^\mu \not{a}\gamma_\mu = -2\not{a}$; p. 12 has $(d-2)\not{a}$. Which is right, and why d?"
- "What is the final $|T|^2$ for $e^-\varphi \to e^-\varphi$, and how do I know it is right?"

In short: every trace follows from the anticommutator, cyclicity and γ_5; the only sign that depends on a convention is that of ϵ^{0123}. The other book's metric is minus ours, and so are its $\not{a}$, $a\cdot b$ and γ_μ: both are right. In dimensional regularization the d multiplies a pole. For $e^-\varphi \to e^-\varphi$, the Dirac equation used before tracing leaves three short traces; the result, (10.29.11), agrees with explicit spinors. The last subsection carries $e^+e^- \to \gamma\gamma$ (LecNote23) the same way to the cross section.

Conventions are Srednicki's: $g = \mathrm{diag}(-1,+1,+1,+1)$, $\{\gamma^\mu,\gamma^\nu\} = -2g^{\mu\nu}$, $\not{a} \equiv \gamma^\mu a_\mu$, $(ab) \equiv a^\mu b_\mu$, so $\not{a}\not{b} + \not{b}\not{a} = -2(ab)$ and $\not{p}\not{p} = -p^2 = m^2$ on shell; $\gamma_5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$, $\gamma_5^2 = 1$, $\{\gamma_5,\gamma^\mu\} = 0$; $\epsilon^{0123} = +1$, so $\epsilon_{0123} = -1$; $\bar\Psi = \Psi^\dagger\beta$, with β equal to γ^0 as a matrix and $\beta\gamma^{\mu\dagger}\beta = \gamma^\mu$. The matrices, the spinors $u_s(\mathbf{p})$ and $\sum_s u_s\bar u_s = -\not{p} + m$ are in §10.26.

Traces without γ_5

Srednicki derives these traces in (47.5)–(47.13) by the steps LecNote17 pp. 10–11 sketch. A trace of an odd number of γ's vanishes ($1 = \gamma_5^2$ before each factor, cyclicity, $\gamma_5\gamma^\mu\gamma_5 = -\gamma^\mu$), and $\mathrm{Tr}[\gamma^\mu\gamma^\nu] = \frac12\mathrm{Tr}\{\gamma^\mu,\gamma^\nu\}$ gives

$$\mathrm{Tr}\,1 = 4, \qquad \mathrm{Tr}[\gamma^\mu\gamma^\nu] = -4g^{\mu\nu}, \qquad \mathrm{Tr}[\not{a}\not{b}] = -4(ab). \tag{10.29.1}$$

For four, $\not{a}$ moves to the right with $\not{a}\not{x} = -\not{x}\not{a} - 2(ax)$; the first step is $\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] = -\mathrm{Tr}[\not{b}\not{a}\not{c}\not{d}] - 2(ab)\,\mathrm{Tr}[\not{c}\not{d}]$, and after the third the trace is back, with a minus sign, so

$$\mathrm{Tr}[\not{a}\not{b}\not{c}\not{d}] = 4\big[(ab)(cd) - (ac)(bd) + (ad)(bc)\big], \tag{10.29.2}$$

p. 11's boxed result. The arrow there expands $\mathrm{Tr}[\not{b}\not{a}\not{c}\not{d}]$ correctly but writes the result under $-\mathrm{Tr}[\not{b}\not{a}\not{c}\not{d}]$, so both of its terms carry the wrong sign. The same steps for any even n give p. 12's "arbitrary even traces",

$$\mathrm{Tr}[\not{a}_1\not{a}_2\cdots\not{a}_n] = \sum_{j=2}^{n}(-1)^{j-1}(a_1a_j)\,\mathrm{Tr}[\not{a}_2\cdots\widehat{\not{a}_j}\cdots\not{a}_n], \tag{10.29.3}$$

where the hat marks the omitted factor; $n=6$ gives 15 terms. A trace also reads backwards: with $C^{-1}\gamma^\mu C=-\gamma^{\mu T}$ (§10.26), $\mathrm{Tr}[\gamma^{\mu_1}\cdots\gamma^{\mu_n}]=(-1)^n\mathrm{Tr}[(\gamma^{\mu_n}\cdots\gamma^{\mu_1})^T]=\mathrm{Tr}[\gamma^{\mu_n}\cdots\gamma^{\mu_1}]$ for even n.

Traces with γ_5, and the sign of ϵ

Since γ_5 is a product of four γ's, it has zero trace with an odd number of γ's. With $(\gamma^0)^2=1$, $\mathrm{Tr}\,\gamma_5=\mathrm{Tr}[\gamma^0\gamma^0\gamma_5]=-\mathrm{Tr}[\gamma^0\gamma_5\gamma^0]=-\mathrm{Tr}\,\gamma_5$, and the same trick with a third index $\rho\neq\mu,\nu$ kills $\mathrm{Tr}[\gamma_5\gamma^\mu\gamma^\nu]$. Hence (p. 12)

$$\mathrm{Tr}\,\gamma_5=0,\qquad \mathrm{Tr}[\gamma_5\not{a}\not{b}]=0,\qquad \mathrm{Tr}[\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma]=-4i\epsilon^{\mu\nu\rho\sigma}. \tag{10.29.4}$$

For the last: a repeated index can be moved next to its partner at the cost of signs, and $(\gamma^\mu)^2=\pm1$ leaves $\mathrm{Tr}[\gamma_5\gamma\gamma]=0$. With four different indices the trace is antisymmetric, so it is a multiple of $\epsilon^{\mu\nu\rho\sigma}$, fixed by $\gamma^0\gamma^1\gamma^2\gamma^3=-i\gamma_5$: $\mathrm{Tr}[\gamma_5\gamma^0\gamma^1\gamma^2\gamma^3]=-4i$. This is $-4i\epsilon^{0123}$ only with Srednicki's $\epsilon^{0123}=+1$, the convention LecNote17 p. 9's γ_5 needs too (§10.26). So $\mathrm{Tr}[\gamma_5\not{a}\not{b}\not{c}\not{d}]=-4i\epsilon^{\mu\nu\rho\sigma}a_\mu b_\nu c_\rho d_\sigma$. Since $\det g=-1$, contracting two ϵ's costs a minus sign: $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\beta}=-6\,\delta^\sigma{}_\beta$ and $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\sigma}=-24$.

Contractions, in d dimensions and in the other metric

With $\gamma_\mu=g_{\mu\nu}\gamma^\nu$ and $g^\mu{}_\mu=d$, moving γ_μ to the left one step at a time, e.g. $\gamma^\mu\not{a}\gamma_\mu=\gamma^\mu(-\gamma_\mu\not{a}-2a_\mu)=(d-2)\not{a}$, gives (p. 12, plus one more)

$$\begin{aligned}\gamma^\mu\gamma_\mu&=-d,\qquad &\gamma^\mu\not{a}\gamma_\mu&=(d-2)\,\not{a},\\ \gamma^\mu\not{a}\not{b}\gamma_\mu&=4(ab)-(d-4)\,\not{a}\not{b},\qquad &\gamma^\mu\not{a}\not{b}\not{c}\gamma_\mu&=2\,\not{c}\not{b}\not{a}+(d-4)\,\not{a}\not{b}\not{c}.\end{aligned} \tag{10.29.5}$$

Only the anticommutator is used, so these hold in any d (checked with explicit Clifford matrices in $d=6$, 8). That is p. 12's "need for dim-reg": loop integrals are done in $d=4-\epsilon$ (§10.18) with the γ's in the numerator beside the pole, and $d-2=2-\epsilon$ times c/ϵ is $2c/\epsilon-c$. The four-dimensional 2 would lose the finite $-c$.

The other book's $\eta=\mathrm{diag}(+1,-1,-1,-1)=-g$ and $\{\gamma^\mu,\gamma^\nu\}=2\eta^{\mu\nu}$ allow the same γ^μ, but its $\not{a}$, $a\cdot b$ and γ_μ are minus ours: (10.29.5) at $d=4$ reads 4, $-2\not{a}$, $4a\cdot b$, $-2\not{c}\not{b}\not{a}$ there, and $\mathrm{Tr}[\not{a}\not{b}]=4a\cdot b$, while (10.29.2) keeps its form.

One spin sum done to the end: $e^-\varphi\to e^-\varphi$

Take $e^-(p)+\varphi(k)\to e^-(p')+\varphi(k')$ at tree level in Yukawa theory, $\mathcal{L}_1=g\varphi\bar\Psi\Psi$, with fermion mass m, scalar mass M (LecNote20's m_φ), $s=-(p+k)^2$, $t=-(p-p')^2$, $u=-(p-k')^2$, $s+t+u=2m^2+2M^2$. The rules of §10.28 give the s- and u-channel diagrams (LecNote20 pp. 2, 4),

$$T=g^2\,\bar u_{s'}(\mathbf{p}')\Big[\frac{-\not{p}-\not{k}+m}{m^2-s}+\frac{-\not{p}+\not{k}'+m}{m^2-u}\Big]u_s(\mathbf{p})\equiv g^2\,\bar u' A u. \tag{10.29.6}$$

For $M < 2m$ (φ cannot decay to e^+e^-) the physical region has $s \geq (m+M)^2 > m^2$ and $u \leq (m-M)^2 < m^2$, so the $i\epsilon$'s are dropped.

Dirac equation first. $(\not{p}+m)u = 0$ turns $-\not{p}$ into $+m$ next to u, so the numerators become $2m - \not{k}$ and $2m + \not{k}'$; and $\bar{u}'(\not{k}' - \not{k})u = \bar{u}'(\not{p} - \not{p}')u = 0$ by momentum conservation and $\bar{u}'(\not{p}' + m) = 0$. So

$$T = g^2\,\bar{u}'(a + b\not{k})u,$$
$$a = 2m\Big(\frac{1}{m^2-s} + \frac{1}{m^2-u}\Big), \qquad b = \frac{1}{m^2-u} - \frac{1}{m^2-s}. \tag{10.29.7}$$

Conjugate and sum. For any 4×4 matrix A, $(\bar{u}'Au)^* = u^\dagger A^\dagger \beta u' = \bar{u}\bar{A}u'$ with $\bar{A} \equiv \beta A^\dagger \beta$. Since $\beta\gamma^{\mu\dagger}\beta = \gamma^\mu$ and the coefficients are real, $\bar{A} = A$ (LecNote20 p. 5, where A^* stands for $A^\dagger$). With the spin sums,

$$\sum_{s,s'} |T|^2 = g^4 \sum_{s,s'} \bar{u}'Au\,\bar{u}Au' = g^4\,\mathrm{Tr}\big[(-\not{p}'+m)\,A\,(-\not{p}+m)\,A\big]. \tag{10.29.8}$$

LecNote20 p. 6 has $(-\not{p}' - m)$ for the first factor (so $(\not{p}' + m)$ on the next line), against the spin sum two lines above it, and $-\not{k}'$ in the second numerator, against p. 4; both are slips, and each changes the value. A third, $\not{p}\not{p} \to p^2$ in its arrow, should read $\not{p}\not{p} = -p^2 = m^2$; the on-page notes there correct all three.

Three traces. Both A's in (10.29.8) stand between on-shell spinors, so $A = a + b\not{k}$ may be used. Odd traces drop, and (10.29.1), (10.29.2) and $k^2 = -M^2$ give

$$\begin{aligned}
\mathrm{Tr}[(-\not{p}'+m)(-\not{p}+m)] &= 4\big(m^2 - (pp')\big),\\
\mathrm{Tr}[(-\not{p}'+m)\not{k}(-\not{p}+m)] &= 4m\big((pk) + (p'k)\big),\\
\mathrm{Tr}[(-\not{p}'+m)\not{k}(-\not{p}+m)\not{k}] &= 8(pk)(p'k) + 4M^2\big((pp') + m^2\big),
\end{aligned} \tag{10.29.9}$$

and $\mathrm{Tr}[(-\not{p}'+m)(-\not{p}+m)\not{k}]$ equals the second. On shell $2(pk) = m^2 + M^2 - s$, $2(p'k) = u - m^2 - M^2$ and $2(pp') = t - 2m^2$, so

$$\sum_{s,s'} |T|^2 = 2g^4\Big[a^2(4m^2 - t) + 2mab\,(u-s) + b^2\big(M^4 - (m^2-s)(m^2-u)\big)\Big]. \tag{10.29.10}$$

Inserting a, b and t, and averaging over the two spin states of the incoming electron (φ has none),

$$\begin{aligned}
\langle|T|^2\rangle \equiv \frac{1}{2}\sum_{s,s'} |T|^2 = g^4\Big[&(4m^2 - M^2)^2\Big(\frac{1}{m^2-s} + \frac{1}{m^2-u}\Big)^2\\
&- 16m^2\Big(\frac{1}{m^2-s} + \frac{1}{m^2-u}\Big) - \frac{(s-u)^2 + 4M^4}{(m^2-s)(m^2-u)}\Big].
\end{aligned} \tag{10.29.11}$$

Checks. LecNote20 p. 6's route, (10.29.8) with the original numerators and $\not{p}\not{p} = m^2$ used inside the traces (the page writes p^2), gives the same. (10.29.11) is symmetric under $s \leftrightarrow u$, as it must be: $k \leftrightarrow -k'$ swaps the two terms of (10.29.6). At $m = M = 0$ it is $g^4(2 - s/u - u/s) = -g^4(s-u)^2/(su)$. It vanishes at threshold, $s = (m+M)^2$, $u = (m-M)^2$: with both particles at rest, $\not{k}u = -M\gamma^0 u = -Mu$, and there $a - bM = 0$. Elsewhere in the physical region it is positive. With explicit spinors $u_s(\mathbf{p}) = \left(\sqrt{-p\cdot\sigma}\,\xi_s, \sqrt{-p\cdot\bar\sigma}\,\xi_s\right)$ (§10.26), the four $T_{ss'}$ computed one by one reproduce (10.29.8) and (10.29.11) at random physical points to rounding error. To test your own algebra: $m = M = 1$, $s = 9$, $u = -1$ ($t = -4$) give $\langle|T|^2\rangle = \frac{113}{64}\,g^4$.

A Second Spin Sum: $e^+e^- \to \gamma\gamma$

LecNote23 pp. 1–3 set up $e^-(p_1) + e^+(p_2) \to \gamma(k_1) + \gamma(k_2)$ and stop inside the first of four traces; p. 4 puts a $\frac{1}{2}$ for the identical photons into the phase space. "How do the traces come out (Srednicki gives only the results, (59.22)–(59.25)), and what are $d\sigma/d\Omega$ and σ?" In short: no four-γ trace is needed, and the $\frac{1}{2}$ goes into σ only.

Take Srednicki's amplitude (59.1) (LecNote23 pp. 1–2; Dirac lines as in §10.28), with $t = -(p_1 - k_1)^2$, $u = -(p_1 - k_2)^2$ (59.2), $N_t \equiv -\not{p}_1 + \not{k}_1 + m$, $N_u \equiv -\not{p}_1 + \not{k}_2 + m$. Spin sums as in (10.29.8) and polarization sums replaced by $g^{\mu\nu}$ (for the full amplitude only: §10.30, Srednicki (59.10)–(59.17)) give his (59.18), $\langle|T|^2\rangle \equiv \frac{1}{4}\sum|T|^2 = e^4\sum_{a,b=t,u}\Phi_{ab}/[(m^2-a)(m^2-b)]$, with the four traces Φ_{ab} of (59.19).

The tt trace. The contractions on p. 3, (10.29.5) at $d = 4$, make Φ_{tt} a quarter of that page's last trace, $\Phi_{tt} = \mathrm{Tr}[N_t(\not{p}_1 + 2m)N_t(\not{p}_2 - 2m)]$. Write $N_t = (m - \not{p}_1) + \not{k}_1$. As $\not{p}_1\not{p}_1 = m^2$, $(m - \not{p}_1)(\not{p}_1 + 2m) = (\not{p}_1 + 2m)(m - \not{p}_1) = m(m - \not{p}_1)$ and $(m - \not{p}_1)^2 = 2m(m - \not{p}_1)$, so the cross terms give $m\{m - \not{p}_1, \not{k}_1\} = 2m^2\not{k}_1 + m(t - m^2)$; as $\not{k}_1\not{k}_1 = 0$, $\not{k}_1(\not{p}_1 + 2m)\not{k}_1 = -2(p_1k_1)\not{k}_1 = (m^2 - t)\not{k}_1$. In all,

$$N_t(\not{p}_1 + 2m)N_t = m(m^2 + t) - 2m^2\not{p}_1 + (3m^2 - t)\,\not{k}_1, \tag{10.29.12}$$

so (10.29.1) suffices: $\Phi_{tt} = -8m^2(m^2+t) + 8m^2(p_1p_2) - 4(3m^2 - t)(p_2k_1)$, which (59.21) turns into (59.22), $\Phi_{tt} = 2[tu - m^2(3t+u) - m^4]$; $k_1 \leftrightarrow k_2$ gives (59.24).

The interference and the sum. In $\Phi_{tu} = \frac{1}{4}\mathrm{Tr}[\gamma^\nu N_t\gamma^\mu(-\not{p}_1 + m)\gamma_\nu N_u\gamma_\mu(-\not{p}_2 - m)]$ the contractions interlock. With $N_t = m - \not{q}$, $N_u = m - \not{q}'$, $q \equiv p_1 - k_1$, $q' \equiv p_1 - k_2$, (10.29.5) term by term gives $\gamma^\nu N_t\gamma^\mu(m - \not{p}_1)\gamma_\nu = 2\not{p}_1\gamma^\mu\not{q} - 4m(p_1+q)^\mu + 2m^2\gamma^\mu$. The γ_μ after N_u then contracts with γ^μ or makes $\not{p}_1 + \not{q}$, and again only traces of two γ's remain: $\Phi_{tu} = 8m^4 - 8(qq')(p_1p_2) + 4m^2[(p_1q) - (p_2q') + (p_1+q)\cdot(q'-p_2)]$. On shell $(qq') = -m^2$, $2(p_1q) = -t - m^2$, $2(p_2q') = u + m^2$ and $(p_1+q)\cdot(q'-p_2) = (2p_1 - k_1)\cdot(k_1 - 2p_2) = s - 4m^2$, so $\Phi_{tu} = 2m^2(s - 4m^2) = \Phi_{ut}$: (59.23), (59.25). In $x \equiv m^2 - t$ and $y \equiv m^2 - u$ ($s = x + y$), (59.18) then

collects into

$$\langle|T|^2\rangle = 2e^4\Big[\frac{y}{x}+\frac{x}{y}+4m^2\Big(\frac{1}{x}+\frac{1}{y}\Big)-4m^4\Big(\frac{1}{x}+\frac{1}{y}\Big)^2\Big], \tag{10.29.13}$$

the form of p. 3's note E3: symmetric under $x \leftrightarrow y$, as two identical photons require, and $2e^4(u/t+t/u)$ at $m=0$, Srednicki's (60.33) ($s_{13}=t$, $s_{14}=u$). To test your algebra: $m=1$, $s=6$, $t=-1$ give $\Phi_{tt}=16$, $\Phi_{uu}=24$, $\Phi_{tu}=4$, $\langle|T|^2\rangle=\frac{13}{2}e^4$.

The cross section. In the CM frame, with θ the angle between $\mathbf{p}_1$ and $\mathbf{k}_1$ and $\beta=(1-4m^2/s)^{1/2}$, $x,y=\frac{s}{2}(1\mp\beta\cos\theta)>m^2$ (no pole). The flux factor $4|\mathbf{p}_1|\sqrt{s}=2\beta s$, $d\text{LIPS}_2=d\Omega/32\pi^2$ (10.13.1) and $\alpha=e^2/4\pi$ give

$$\frac{d\sigma}{d\Omega}=\frac{\langle|T|^2\rangle}{64\pi^2\beta s}=\frac{\alpha^2}{\beta s}\,\frac{1+2\beta^2\sin^2\theta-\beta^4-\beta^4\sin^4\theta}{(1-\beta^2\cos^2\theta)^2}, \tag{10.29.14}$$

with no $\frac{1}{2}$: as in §10.13, it counts photons in a detector. The $\frac{1}{2}$ of p. 4 goes into $\sigma=\frac{1}{2}\int d\Omega\,d\sigma/d\Omega$; with $d\Omega=(4\pi/\beta s)\,dx$ over $\frac{s}{2}(1\mp\beta)$, $\int dx/x=\ln\frac{1+\beta}{1-\beta}\equiv\ell$, $\int dx/x^2=\beta/m^2$ and $\int dx/xy=2\ell/s$ (the same with y) give

$$\sigma=\frac{\pi\alpha^2}{\beta^2 s}\Big[(3-\beta^4)\,\ell-2\beta(2-\beta^2)\Big]\;\to\;\frac{2\pi\alpha^2}{s}\Big[\ln\frac{s}{m^2}-1\Big]\quad(s\gg m^2), \tag{10.29.15}$$

24.4 nb at $\sqrt{s}=10$ GeV ($\alpha=1/137.036$, (10.13.8)). The logarithm is the forward peak, $d\sigma/d\Omega\approx(2\alpha^2/s)/(\theta^2+4m^2/s)$: the $1/t$ of (60.33) cut off at $\theta\approx 2m/\sqrt{s}$.